

ECOALERT - LORA BASED SMART WILDLIFE DETECTION AND ALERT SYSTEM

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Vision: Progress through the growing knowledge of Electronics and Communication technology.

Mission: "M1: To provide students state-of-the-art academic ambience for quality education. M2: Facilitate interdisciplinary learning, industry interactions and research initiatives for long-term career development goals. M3: To educate and mentor students on professional responsibilities, ethical values, and lifelong learning".

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- Animal-train collisions cause wildlife fatalities and railway infrastructure damage.
- Existing solutions lack real-time detection and effective early warnings.
- A smart wildlife detection and alert system using MobileNet V2 CNN.
- Combines image processing, LoRa, GSM, GPS, and IR sensors for real-time alerts.
- Prevents collisions, improves railway safety, and aids in wildlife conservation.

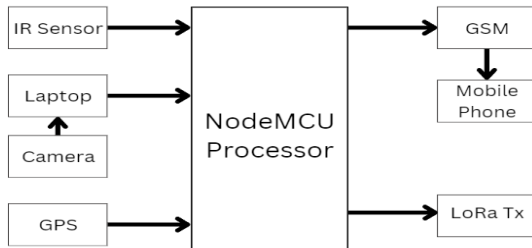
- **Frequent Wildlife Collisions:** Traditional methods like fencing, signboards, and manual patrolling are ineffective in preventing animal-train collisions.
- **Limited Visibility Detection:** Accidents occur in low-light or dense forest areas, where real-time wildlife detection is challenging.
- **Lack of Instant Alerts:** The absence of a reliable, long-range, and low-power communication system delays preventive action.

- Reduce collisions by preventing animal-train accidents.
- Improve safety by minimizing train delays and infrastructure damage.
- Detect animals in real-time using CNN-based image processing.
- Send instant alerts to train operators for quick action.
- Protect wildlife by preserving natural habitats and migration paths.
- Implement a scalable system for widespread railway deployment.

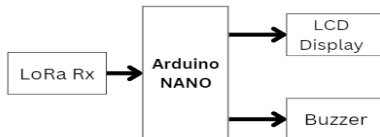
Author	Title	Contribution
Ranjith, A., Vijayaragavan, S. P., & Muthukumaran, N.(2022)	An IoT-Based Monitoring System to Detect Animals on Railway Tracks Using Deep Learning Neural Networks	Developed an IoT-based system using deep learning to detect animals on railway tracks, reducing collision risks.
Sabnis, O. V., & L.R. (2019)	A Novel Object Detection System for Improving Safety at Unmanned Railway Crossings	Proposed an object detection system for railway crossings to enhance safety using machine learning techniques.
Gómez, A., Salazar, A., & Vargas, F. & N. Kumar (2017)	Towards Automatic Wild Animal Monitoring: Identification of Animal Species in Camera-Trap Images Using Very Deep CNNs	Applied deep CNNs to automate species identification in camera-trap images, improving wildlife monitoring efficiency.
Mamat, N., Othman, M. F., & Yakub, F. (2022)	Animal Intrusion Detection in Farming Area using YOLOv5 Approach	Used YOLOv5 for real-time animal intrusion detection in agricultural areas, enhancing farm security.
Ghodhbane, R. M., Hernandez, A., & Janfaoui, S. (2024)	A Pressure and Temperature Wireless Sensing Network Communicating with LoRa Modulation	Designed a LoRa-based wireless sensing network for remote monitoring of pressure and temperature.

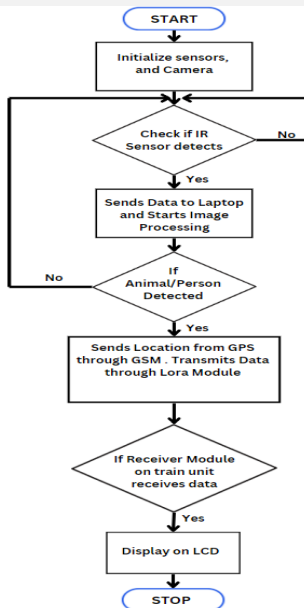
- **Hardware Development:** IR sensor detects obstacles, the camera captures images, and NodeMCU controls the system. LoRa GSM send alerts, and the train unit receives warnings.
- **System Activation:** When the train enters a 10 km range, the LoRa transmitter sends a signal to the train unit.
- **Wildlife Detection:** The IR sensor triggers the camera, and MobileNet V2 CNN processes images to identify animals.
- **Alert Transmission:** If an animal is detected, LoRa sends data to the train, while GSM GPS notify railway authorities.
- **Train Warning System:** A buzzer and LCD display alert the driver in real-time to take preventive action.

Transmitter Unit



Receiver Unit





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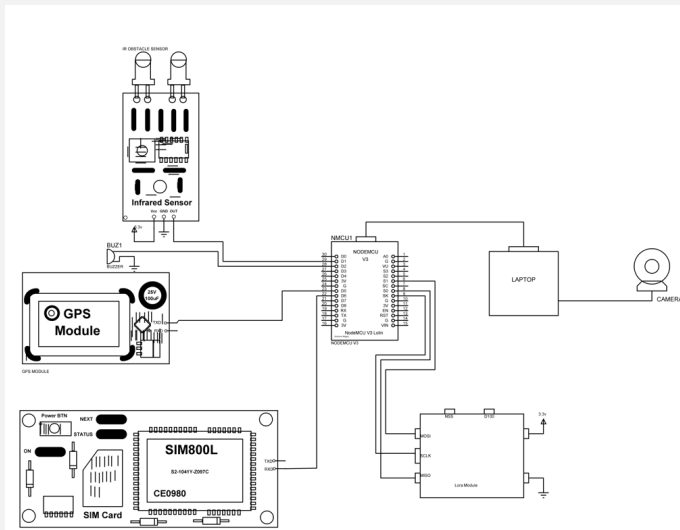
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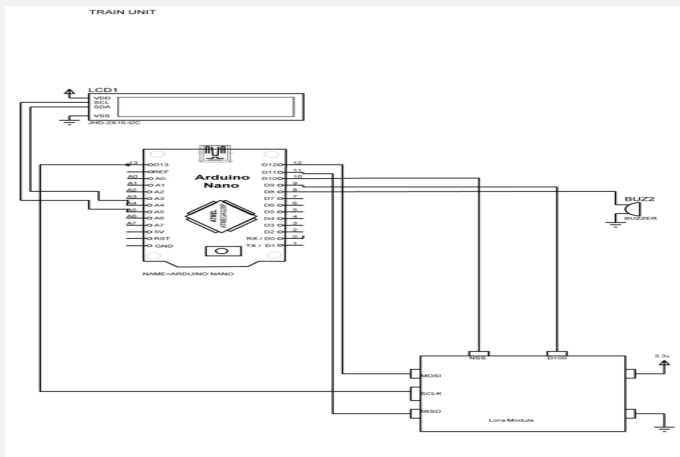
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- Camera: Captures real-time images of railway tracks for wildlife detection.
- IR Sensor: Detects obstacles and triggers the camera for image capture.
- NodeMCU: Controls the entire system and manages communication between components.
- LoRa Module: Enables long-range, low-power wireless communication between the detection unit and the train.
- GSM Module: Sends alerts to railway authorities and train operators.
- GPS Module: Provides location data for accurate tracking and alerting.
- LCD Display: Shows warning messages to train drivers in real-time.
- Buzzer: Generates an audible alert to warn train operators about detected wildlife.

1. System Initialization Sensor Setup

- Integrated IR sensor for obstacle detection, GPS for location tracking, and GSM/LoRa for communication.
- Ensured all components activate upon system startup and function in real-time.

2.CNN-Based Wildlife Detection

- IR sensor detects an obstacle, triggering the camera to capture an image.
- The image is preprocessed (resized, filtered) before being fed into the Convolutional Neural Network (CNN).
- The CNN extracts key features (edges, textures, and shapes) from the image using convolution layers.
- Pooling layers reduce dimensionality while retaining important details.
- The extracted features are passed through fully connected layers, where the softmax activation function classifies the object (elephant, cow, dog, etc.).
- If the detected object matches a predefined wildlife category, the alert system is activated.

3.LoRa GSM-Based Alert Transmission

- LoRa Module Transmission: The transmitter unit sends the alert signal over long distances using a LoRa antenna.
- LoRa Receiver in Train: The train's LoRa module continuously listens for incoming signals and activates an onboard alert system.
- GSM SMS Alert: The system also sends an SMS alert to railway authorities with location details for quick action.

4. Software Simulation Prototype Testing

- Conducted simulations using Arduino Nano, Proteus, and Arduino IDE for hardware validation.
- Trained and tested the CNN model using Anaconda Navigator and Spyder to ensure accurate image classification.
- Verified LoRa transmission range and real-time response of the system in prototype testing.

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Figure 1: System

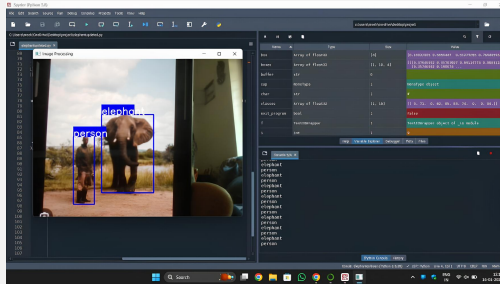


Figure 2: CNN detecting an elephant and a person together

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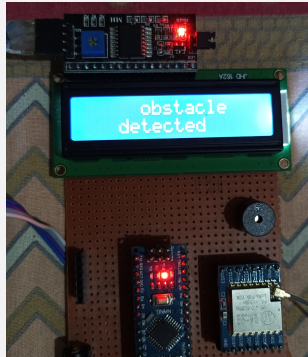


Figure 3: LCD display showing alert message

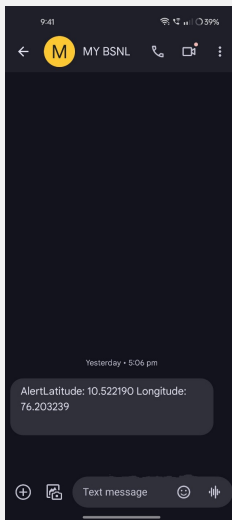


Figure 4: Alert received through SMS with location of the detected place

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Autonomous Train Braking System- Implementing automated braking to enhance safety by minimizing human response time.

Blockchain for Secure Communication – Using blockchain to secure alerts and prevent cyber threats.

Integration with Railway Traffic Management Systems – Linking with railway control centers for real-time monitoring and route adjustments.

AI-Powered Threat Classification – Utilizing AI to differentiate between wildlife, humans, and objects for more accurate threat detection.

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